cp_10c

1

Sum of N Terms in Power Series | 35/35

Easy

Given two numbers `X` and `N`, write a program to print the **sum** of `N` terms in the given series.

Series:

```
X^1, X^3, X^5, ... N terms
```

Terms in the given series:

Term Number	Term	Power
First Term	X^1	1
Second Term	X^3	3
Third Term	X^5	5
..	..	..
Nth Term	X^{2N-1}	2N-1

Input

The first line of input contains an integer representing `X`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the **sum** of the series from the **first term** to the **Nth term**.

Explanation

For example, if the given numbers are `X = 2` and `N = 6`,

- Number of terms is **6**.
- The terms from **First Term** to **Sixth Term** in the given series are **2** (2^1), **8** (2^3), **32** (2^5), **128** (2^7), **512** (2^9) and **2048** (2^{11}

  Terms in the given series:

Term Number	Term (X = 2)	Power
First Term	2^1 (2)	1
Second Term	2^3 (8)	3
Third Term	2^5 (32)	5
Fourth Term	2^7 (128)	7
Fifth Term	2^9 (512)	9
Sixth Term	2^{11} (2048)	11

- Sum of the terms from the **First Term** to the **Sixth Term** is **2730**. (`2 + 8 + 32 + 128 + 512 + 2048 = 2730`)

code:

```
x=int(input())
n=int(input())
sum=0
for i in range(1,n+1):
    row=str(x)
    power_of=int(row)**((2*i)-1)
    sum=sum+power_of
print(sum)
```

2

Sum of N Terms in Power Series - 2 | 30/30

Easy

Given two numbers `X` and `N`, write a program to print the **sum** of `N` terms in the given series.

Series:

```
X^2, X^4, X^6, ... N terms
```

Terms in the given series:

Term Number	Term	Power
First Term	X^2	2
Second Term	X^4	4
Third Term	X^6	6
..	..	..
Nth Term	X^{2N}	2N

Input

The first line of input contains an integer representing `X`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the **sum** of the series from the **first term** to the **Nth term**.

Explanation

For example, if the given numbers are `X = 3` and `N = 4`,

- Number of terms is **4**.
- The terms from **First Term** to **Fourth Term** in the given series are **9** (3^2), **81** (3^4), **729** (3^6) and **6561** (3^8).

  Terms in the given series:

Term Number	Term (X = 3)	Power
First Term	3^2 (9)	2
Second Term	3^4 (81)	4
Third Term	3^6 (729)	6
Fourth Term	3^8 (6561)	8

- Sum of the terms from the **First Term** to the **Fourth Term** is **7380**. (`9 + 81 + 729 + 6561 = 7380`)

The output should be **7380**.

code:

```
x=int(input())
n=int(input())
total=0
for i in range(1,n+1):
    power=x**(2*i)
    total=total+power
print(total)
```

3

Sum of N Terms in Power Series - 3 | 30/30

Solved

Easy

Given two numbers `X` and `N`, write a program to print the **sum** of `N` terms in the given series.

Series:

$X^2, -X^4, X^6, -X^8, \ldots$ N terms

Terms in the given series:

Term Number	Term	Power
First Term	X^2	2
Second Term	$-(X^4)$	4
Third Term	X^6	6
Fourth Term	$-(X^8)$	8
..	..	..

Input

The first line of input contains an integer representing `X`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the **sum** of the series from the **first term** to the **Nth term**.

Explanation

For example, if the given numbers are `X = 2` and `N = 6`,

- Number of terms is **6**.
- The terms from the **First Term** to the **Sixth Term** in the given series are **4** (2^2), **-16** ($-(2^4)$), **64** (2^6), **-256** ($-(2^8)$), **1024** (2^{10}) and **-4096** ($-(2^{12})$).

  Terms in the given series:

Term Number	Term (X = 2)	Power
First Term	2^2 (4)	2
Second Term	$-(2^4)$ (-16)	4
Third Term	2^6 (64)	6
Fourth Term	$-(2^8)$ (-256)	8
Fifth Term	2^{10} (1024)	10
Sixth Term	$-(2^{12})$ (-4096)	12

- Sum of the terms from the **first term** to the **sixth term** is **-3276**. (`4 + (-16) + 64 + (-256) + 1024 + (-4096) = -3276`)

code:

```
x=int(input())
n=int(input())
sum=0
for i in range(1,n+1):
    power=2*i
    row=x**power
    if power%4==0:
        row=-(row)
        sum=sum+row
    else:
        row=x**power
        sum=sum+row
print(sum)
```

4

Right Angled Triangle - 5 | 40/40

Easy

Given a number `N` , write a program to print the **Right Angled Triangle** of `N` rows using stars (`*`).

```
  *
 **
***
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) on **4** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    row=(" "*(n-1))+("*"*i)
    n=n-1
    print(row)
```

5

Right Angled Triangle - 6 | 35/35

Easy

Given a number `N` , write a program to print the **Right Angled Triangle** of `N` rows using stars (`*`).

```
    *
   * *
  * * *
 * * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 6` ,

- Print the stars (`*`) on **6** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    row=(("  "*(n-1))+("* ")*i)
```

```
    n=n-1
    print(row)
```

6

Right Angled Triangle - 7 | 40/40 ● Solved

Easy

Given a number `N` , write a program to print the **Right Angled Triangle** of `N` rows using stars (`*`) and hashes (`#`).
The last row contains hashes (`#`) and the remaining rows contain stars (`*`).

```
   *
  **
 ***
####
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) and hashes (`#`) as a **Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the stars (`*`) and hashes (`#`) on **5** rows as shown in the given image.
  - The first **4** rows contain stars (`*`).
  - The **5th** row contains hashes (`#`).

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    if i!=n1:
        row=(" "*(n-1))+("*"*i)
        print(row)
    else:
       row=(" "*(n-1))+"#"*n1
       print(row)
    n=n-1
```

7

Pyramid | 35/35

Easy

Given a number `N` , write a program to print a **Pyramid** of `N` rows using stars (`*`).

```
    *
  * * *
* * * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) on **4** rows as shown in the given image.

code:

```
n=int(input())
for i in  range(1,n+1):
    power=(2*i)-1
    no_of_stars="* "*power
    row=("  "*(n-1))+(no_of_stars)
    n=n-1
    print(row)
```

8

Pyramid - 2 | 35/35

Easy

Given a number `N` , write a program to print a **Pyramid** of `N` rows using numbers.

```
    1
  2 2 2
3 3 3 3 3
```

Note

There is a space after every number.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing numbers as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- The numbers should be printed on **5** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    str_i=str(i)+" "
    power=(2*i)-1
    row=("  "*(n-1))+(str_i*power)
    n=n-1
    print(row)
```

9

Inverted Right Angled Triangle - 4 | 45/45

Easy

Given a number `N` , write a program to print an **Inverted Right Angled Triangle** of `N` rows using numbers.

```
4444
 333
  22
   1
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing numbers as an **Inverted Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the numbers on **3** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    str_i=str(n)
    row=" "*(i-1)+str_i*n
    n=n-1
    print(row)
```

10

Inverted Right Angled Triangle - 5 | 45/45

Easy

Given a number `N` , write a program to print an **Inverted Right Angled Triangle** of `N` rows using numbers.

```
3 3 3
  2 2
    1
```

Note

There is a space after every number.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing numbers as an **Inverted Right Angled Triangle** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the numbers on **4** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    str_n=str(n)+" "
    star=str_n*n
    row="  "*(i-1)+star
    n=n-1
    print(row)
```

11

Inverted Pyramid | 35/35

Easy

Given a number `N` , write a program to print an **Inverted Pyramid** of `N` rows using stars (`*`).

```
*****
 ***
  *
```

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as an **Inverted Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- The stars (`*`) should be printed on **4** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    row=" "*(i-1)+"*"*((2*n)-1)
    n=n-1
    print(row)
```